

# PLACEMENT HUB

A Placement Management System using Data Structures and Algorithms

2520030128 – Sai Sharan

2520030083 – Saanvi

2520030487 – Sindhu

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DATA STRUCTURES AND ALGORITHMS – 3

# ABSTRACT

**Placement Hub** is a placement management system designed to simplify and organize recruitment activities in educational institutions. It provides a centralized platform for managing student profiles, company details, job opportunities, eligibility criteria, and recruitment drives, reducing the effort involved in manual placement management.

The system helps match students with suitable job opportunities based on their academic details, skills, and company requirements. By applying Data Structures and Algorithms (DSA) for efficient searching, sorting, storing, and processing of data, Placement Hub improves placement management and helps students discover relevant career opportunities efficiently.

# PROBLEM STATEMENT

## EXISTING PROBLEMS

- ☐ Placement information is often managed manually.
- ☐ Students find it difficult to track all recruitment drives.
- ☐ Matching eligible students with company requirements takes time.
- ☐ Managing student and company records becomes inefficient as data grows.

## PROPOSED SOLUTION

Develop **Placement Hub**, a centralized platform that stores placement-related information and automatically helps match eligible students with suitable job opportunities using efficient DSA techniques.

## PROPOSED FUNCTIONALITIES

- ✓ Student Registration & Login
- ✓ Student Profile Management
- ✓ Company Profile Management
- ✓ Recruitment Drive Management
- ✓ Eligibility Checking
- ✓ Search Companies & Jobs
- ✓ Sort Students based on CGPA
- ✓ Job Application Tracking
- ✓ Student–Company Matching

# TOOLS AND TECHNOLOGIES

## ➤ **Programming Language**

➤ Java

## ➤ **Database**

➤ MySQL

## ➤ **Frontend**

➤ HTML

➤ CSS

➤ JavaScript

## ➤ **Backend**

➤ Java (JDBC)

## ➤ **DSA Concepts Used**

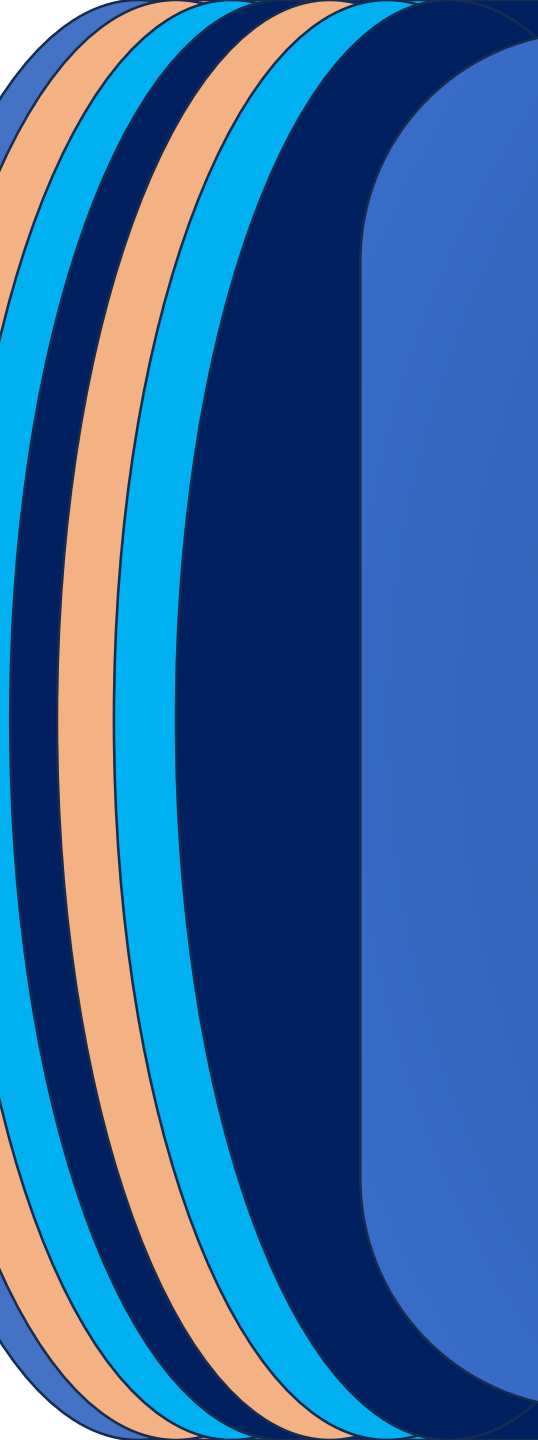
➤ Arrays

➤ Linked Lists

➤ Searching Algorithms

➤ Sorting Algorithms

➤ HashMap (for efficient data retrieval)



## EXPECTED OUTCOME

- Centralized placement management system.
- Faster management of student and company data.
- Easy eligibility verification.
- Quick search and sorting of placement records.
- Better matching between students and companies.
- Improved efficiency and reduced manual work for the placement cell.